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LAB-1 | Output Report

1. 1. Implement Stack data structure using an array as well as a linked list.

(a) **push(element)**: Adds an element into the stack

=>

Using Array:

```
Choose one of the following options:
1. Array Stack
2. Linked Stack
Enter your option: 1
Choose one of the following choice:
1. push(element): Adds an element into the stack
2. pop(): Removes an element from the stack
3. isEmpty(): Checks if the stack is empty
4. isFull(): Checks if the stack is full
5. top(): Gives the element at the top
6. exit
Enter your choice: 1
Enter the element: 1
Pushed element is 1
Enter your choice: 1
Enter the element: 2
Pushed element is 2
Enter your choice: 1
Enter the element: 3
Pushed element is 3
Enter your choice: █
```

Using Linked list:

```

Choose one of the following options:
1. Array Stack
2. Linked Stack
Enter your option: 2
Choose one of the following choice:
1. push(element): Adds an element into the stack
2. pop(): Removes an element from the stack
3. isEmpty(): Checks if the stack is empty
4. isFull(): Checks if the stack is full
5. top(): Gives the element at the top
6. exit
Enter your choice: 1
Enter the element: 5
Element 5 pushed into the stack
Enter your choice: 1
Enter the element: 6
Element 6 pushed into the stack
Enter your choice: 1
Enter the element: 7
Element 7 pushed into the stack
Enter your choice: █

```

(b) **pop()**: Removes an element from the stack

=>

Using Array:

```

Enter your choice: 2
Element popped from the stack is 3
Enter your choice: 2
Element popped from the stack is 2
Enter your choice: █

```

Using Linked list:

```

Enter your choice: 2
Element popped from the stack is 7
Enter your choice: █

```

(c) **isEmpty()**: Checks if the stack is empty

=>

Using Array:

```

Enter your choice: 3
Stack is not empty
Enter your choice: █

```

Using Linked list:

```
Enter your choice: 3  
Stack is not empty  
Enter your choice: █
```

(d) **isFull()**: Checks if the stack is full

=>

Using Array:

```
Stack is not full  
Enter your choice: █
```

Using Linked list:

```
Enter your choice: 4  
Stack is not full  
Enter your choice: █
```

(e) **top()**: Gives the element at the top

=>

Using Array:

```
Enter your choice: 5  
Top element is 1  
Enter your choice: █
```

Using Linked list:

```
Enter your choice: 5  
Top element is 6  
Enter your choice: █
```

- Overall output using array:

```
Choose one of the following options:
1. Array Stack
2. Linked Stack
Enter your option: 1
Choose one of the following choice:
1. push(element): Adds an element into the stack
2. pop(): Removes an element from the stack
3. isEmpty(): Checks if the stack is empty
4. isFull(): Checks if the stack is full
5. top(): Gives the element at the top
6. exit
Enter your choice: 1
Enter the element: 1
Pushed element is 1
Enter your choice: 1
Enter the element: 2
Pushed element is 2
Enter your choice: 1
Enter the element: 3
Pushed element is 3
Enter your choice: 2
Element popped from the stack is 3
Enter your choice: 2
Element popped from the stack is 2
Enter your choice: 3
Stack is not empty
Enter your choice: 4
Stack is not full
Enter your choice: 5
Top element is 1
Enter your choice: 6
```

```
○ dipenkhatri@Dipens-MacBook-Pro build %
```


- Overall output using linked list:

```
Choose one of the following options:
1. Array Stack
2. Linked Stack
Enter your option: 2
Choose one of the following choice:
1. push(element): Adds an element into the stack
2. pop(): Removes an element from the stack
3. isEmpty(): Checks if the stack is empty
4. isFull(): Checks if the stack is full
5. top(): Gives the element at the top
6. exit
Enter your choice: 1
Enter the element: 5
Element 5 pushed into the stack
Enter your choice: 1
Enter the element: 6
Element 6 pushed into the stack
Enter your choice: 1
Enter the element: 7
Element 7 pushed into the stack
Enter your choice: 2
Element popped from the stack is 7
Enter your choice: 3
Stack is not empty
Enter your choice: 4
Stack is not full
Enter your choice: 5
Top element is 6
Enter your choice: 6
○ dipenkhatri@Dipens-MacBook-Pro build %
```